The Neurodharma of Love:

Using Brain Science and Buddhist Wisdom

To Illuminate the Heart

Of Important Relationships

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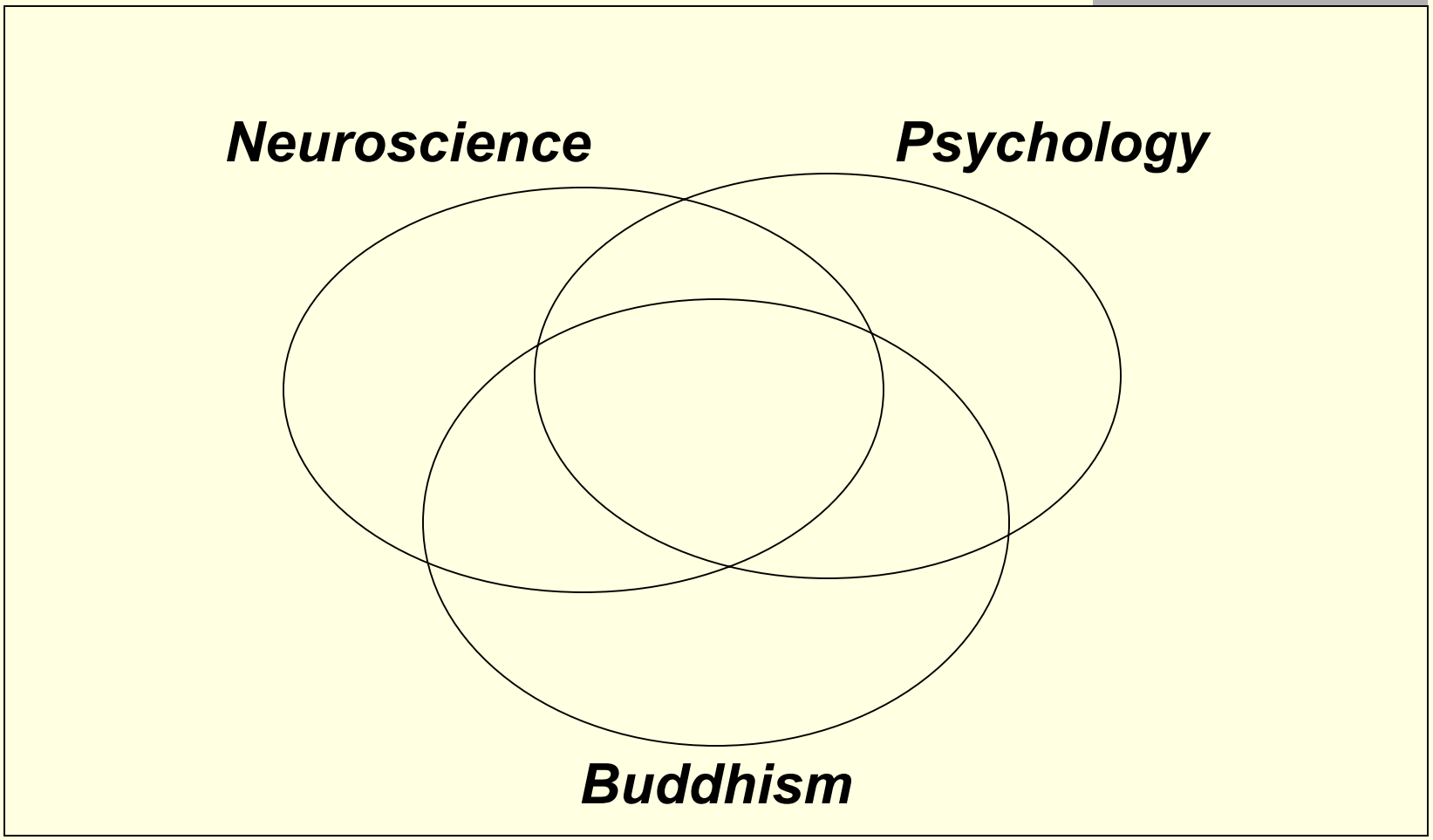
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Plan for the Day

- Context
- How Your Mind Can Change Your Brain
- Balancing Joining and Separation
- Goodness and Virtue in Relationships
- Empathy
- The Evolution of Love
- The Dark Side of Bonding
- Lovingkindness for the Whole Wide World

Common - and Fertile - Ground



Basics of Meditation

- Relax
- Posture that is comfortable and alert
- Simple good will toward yourself
- Awareness of your body
- Focus on something to steady your attention
  - Sensations of breath
  - Phrase like, “May all beings be happy.”
- Accepting whatever passes through awareness, not resisting it or chasing it
- Gently settling into peaceful well-being

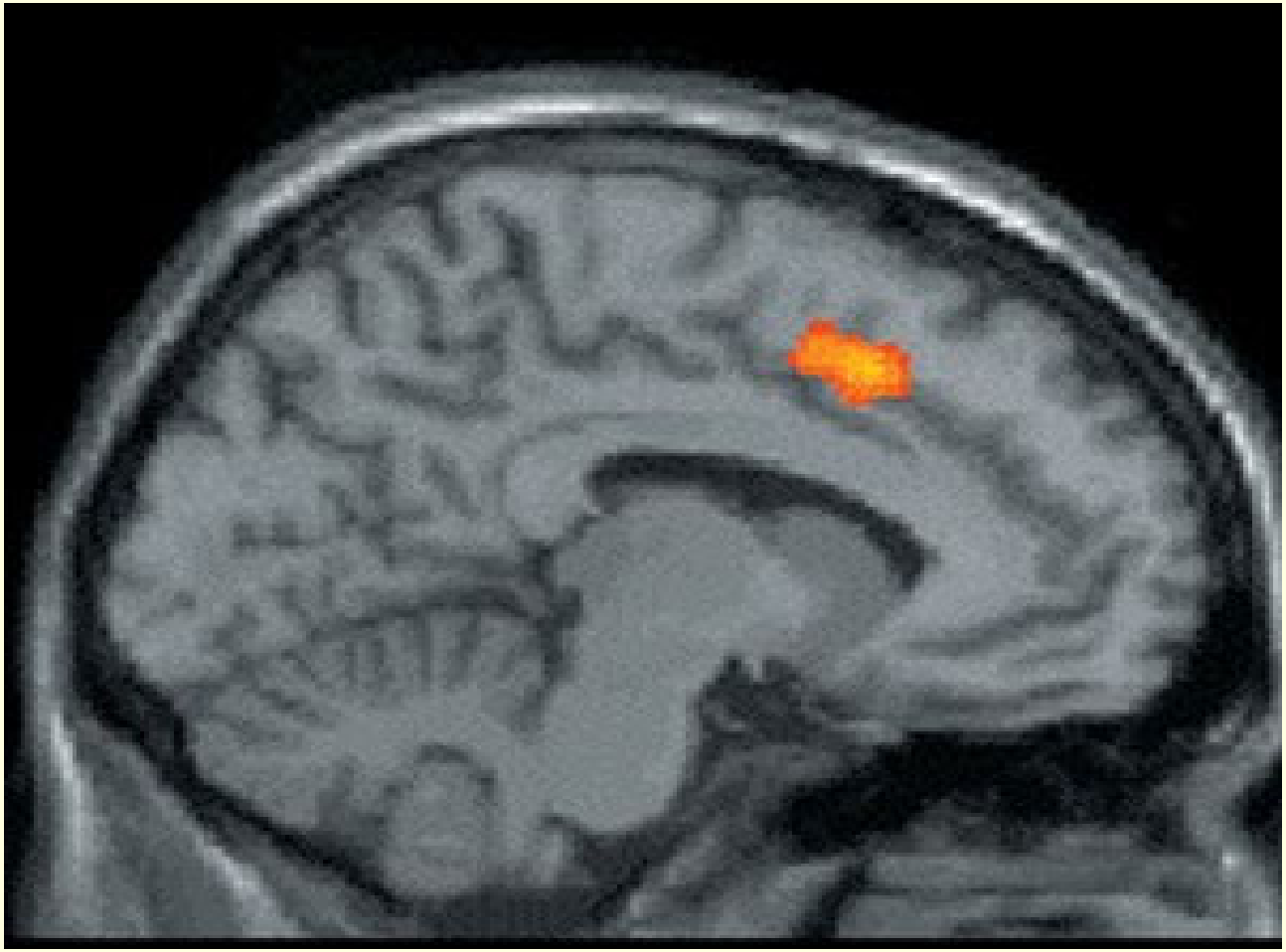
The Union of Mind and Body

- Subjective experience correlates with brain activities.
- Change your experience - and you change your brain, temporarily and then permanently.
- Change your brain - and you change your experience.

Meditation - Immediate Physical Effects

- Lowers the heart rate
- Activates the soothing, “rest-and-digest” parasympathetic nervous system (PNS)
- Settles down the “fight-or-flight,” stress response sympathetic nervous system (SNS)
  - Many health problems of chronic SNS activation
- Changes brainwaves
- Lights up brain regions handling sensory awareness and control of attention

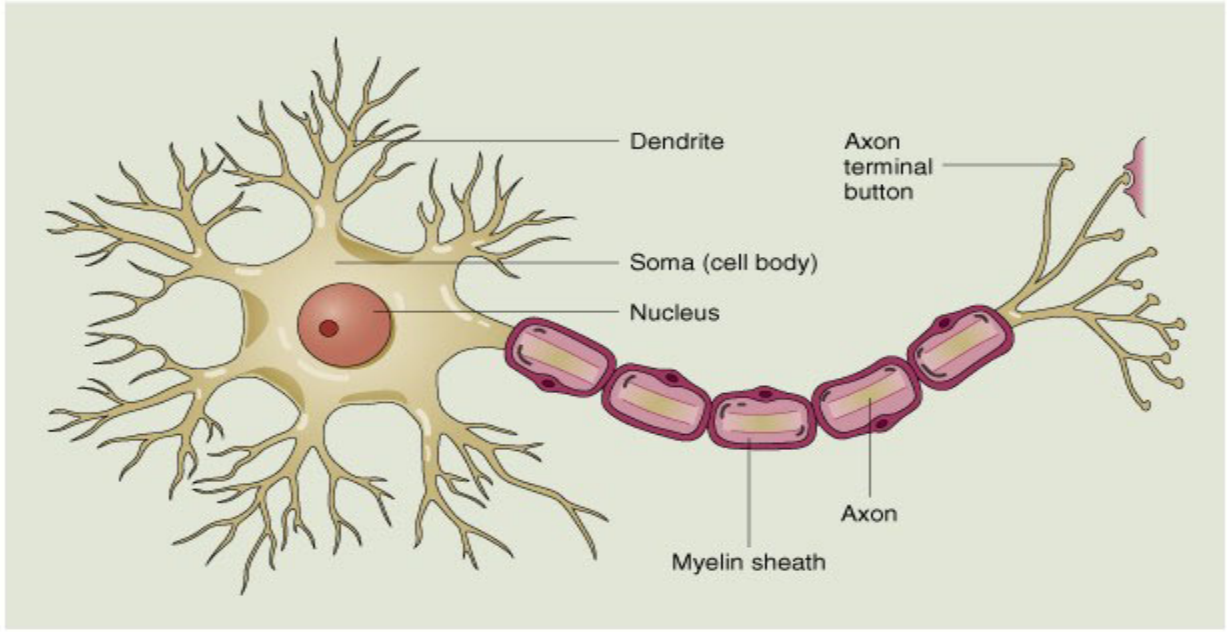
“Ardent, Diligent, Resolute, and Mindful”



Meditation - Long-term Physical Effects

- Dampens reactivity to stress
  - Dampens amygdala responses
  - Sensitizes and “empowers” parasympathetic system
- Thickens brain regions (= makes them stronger):
  - Anterior cingulate (controlling attention, integrating thinking/feeling)
  - Parts of parietal lobes (sensory awareness)
  - Less cortical thinning with aging
- Increases left frontal activation: more positive emotions
- Primes brain to enter states of very fast pulses of coherent neural activity integrating broad regions

One Simple Neuron . . .



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Balancing Joining and Separation

- We want to be our own person, and we want to join with others.
- This balance - closeness and distance, intimacy and autonomy - varies from individual to individual, due to genetic factors, and the effects of personal choices and environmental influences.
- It's natural to be wary of closeness:
  - Most psychological wounds occur in relationships.
  - "Stranger danger" in our evolutionary history
- Individuality and relationship support each other; "fences make for good neighbors."
- Feeling strong helps you enter the depths of relationship.

Feeling Strong

- Relaxed, resting in awareness
- Feeling the strength in awareness itself, never sullied or rattled by what passes through it.
- Sense the vitality in your body
- Recall a time you felt really strong, and sense those feelings again.
- Energy and strength in your breathing . . . in arms and legs . . . in your whole being . . .
- A spacious strength that lets others flow through
- In relationship and at peace
- Relaxed in a spacious world; no need for struggle

Three Pillars of Buddhist Practice

- Sila - Restraint, morality, virtue
- Samadhi - Steadiness of mind, meditative absorption, non-ordinary states of being
- Panna - Wisdom

The point is not that these are "good,"
but that they are effective.

How Virtue Supports Relationships

- Brings peace of mind, “the bliss of blamelessness.”
- Reduces interpersonal wrangles.
- Encourages good behavior in others.
- Teaches us what we can ask for from others.

Goodness and Virtue

- Virtue is the outer expression of inner goodness.
  - Virtue: “top-down”
  - Goodness: “bottom-up
- The ultimate is unbidden, naturally flowing goodness.
- But that’s hard, so we take refuge in virtue.
- The real test of virtue is doing the right thing when we don’t feel like it.
- Virtues are specific and clear; it’s usually pretty clear if we are living by some virtue.

Buddhist Relationship Virtues

- The Five Precepts: Do not kill, steal, lie, create harms through sexuality, or abuse intoxicants to the point of heedlessness.

- Wise Livelihood: Do not trade in weapons, living beings, meat, intoxicants, or poisons.

- Wise Speech: Say only what is true, beneficial, timely, expressed without harshness, and - ideally - what is wanted.

These are framed as aids to practice,
not rules that are a sin to break.

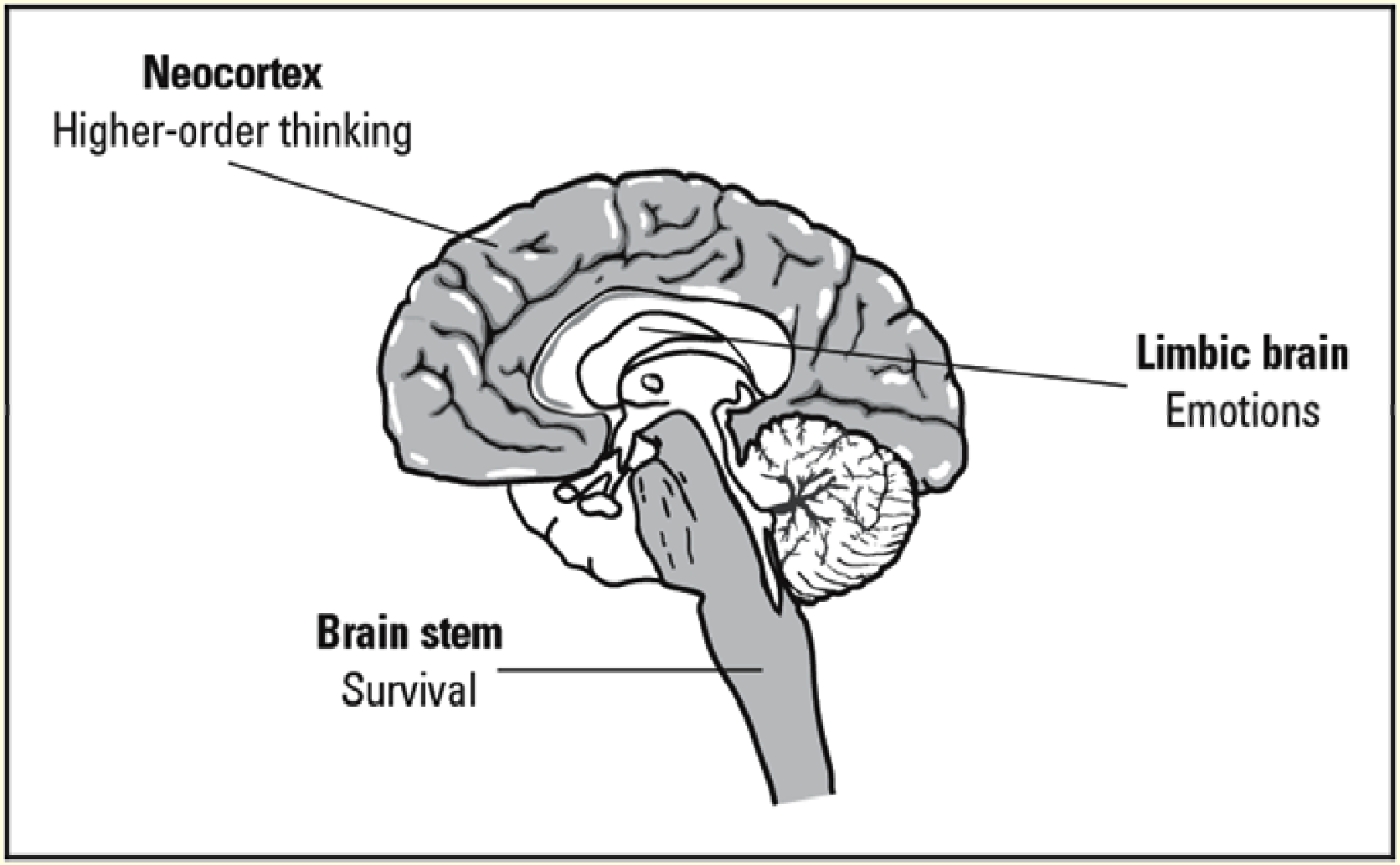
Wise Speech

Speech that is:

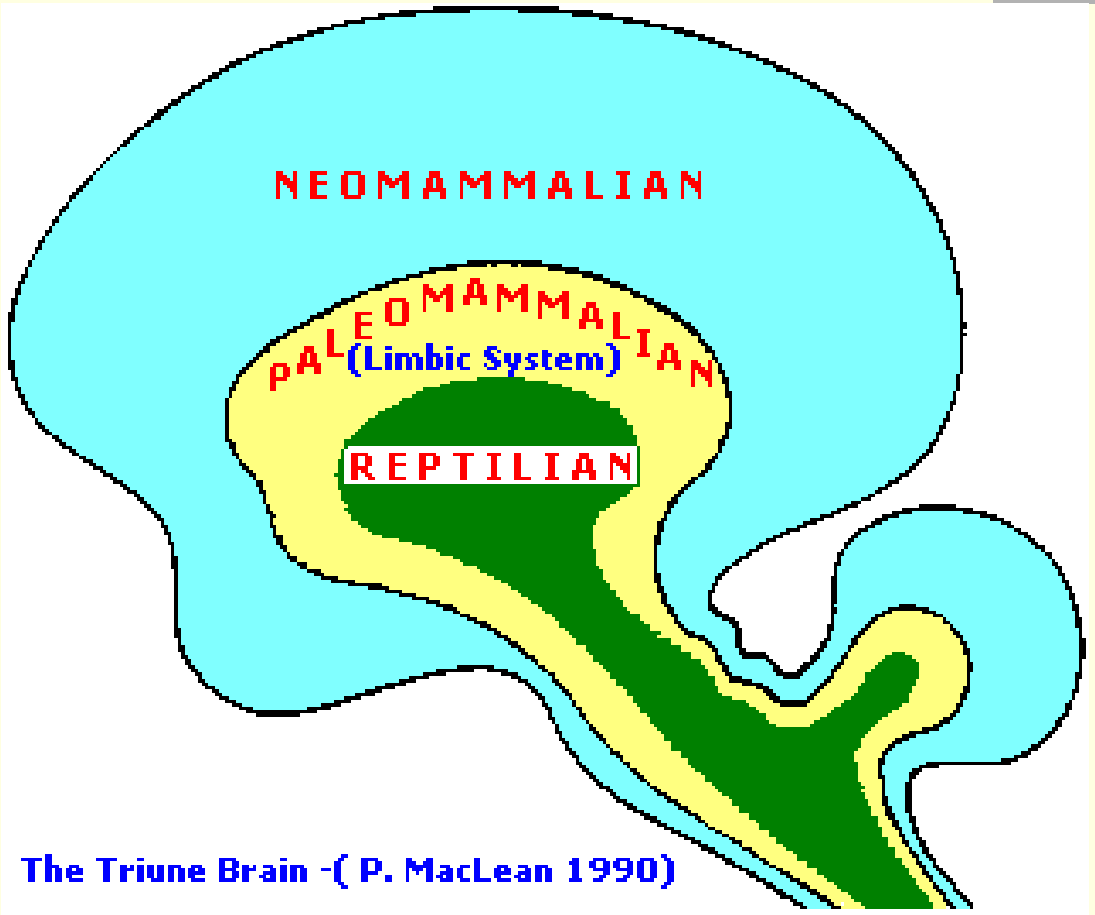
- True
- Beneficial
- Timely
- Expressed without harshness
- Ideally, wanted.

Fundamentally, speech that comes from good will rather than ill will.

Circuits of Emotional Responses



Evolutionary History



The Triune Brain

The Anatomy of Reactivity

- Incoming stimuli processed by limbic system
  - Central switchboard
  - Primed to go negative: anxious combativeness
- Snap judgments [influenced by memory via hippocampus]:
  - Pleasant → Approach (mate with or eat)
  - Unpleasant → Avoid (flee from, fight with, freeze, or appease)
  - Neutral → Ignore, forget, move on
- Reacts before frontal lobes can process perception signals
  - *"Jump first, ask questions later!"*
- Can lead ancient, visceral reactions to hijack modern, reasoning mind (especially with history of trauma)

Points of Intervention

- In your environment:
  - Do whatever you can, that is wise, virtuous, etc.
  - But limited influence on your environment.
  - And only partial effects.
- In your inner world:
  - Much more influence
  - Changes are with you wherever you go
- Working internally with the frontal lobe/limbic circuits:
  - Train limbic to be less reactive
  - Increase top-down control
    - Foundation that prevents limbic reactivity
    - Increase alertness to emotional triggers
    - More powerful control over limbic system

Wholesome Intentions in Relationships

- Evolutionary priming to use tools
- Hard-wired inclination to see ourselves as an agent: a cause, not an effect; a hammer, not a nail
- Intentions require agency to be effective.
- “Be a light unto yourself.”
- Wise Intention is the natural expression of Wise View: We see that we cause effects and must inherit their fruits, that we cannot change the moment that has arrived but can change the moments that follow.

God grant me the serenity •

to accept the things I cannot change; •

courage to change the things I can;•

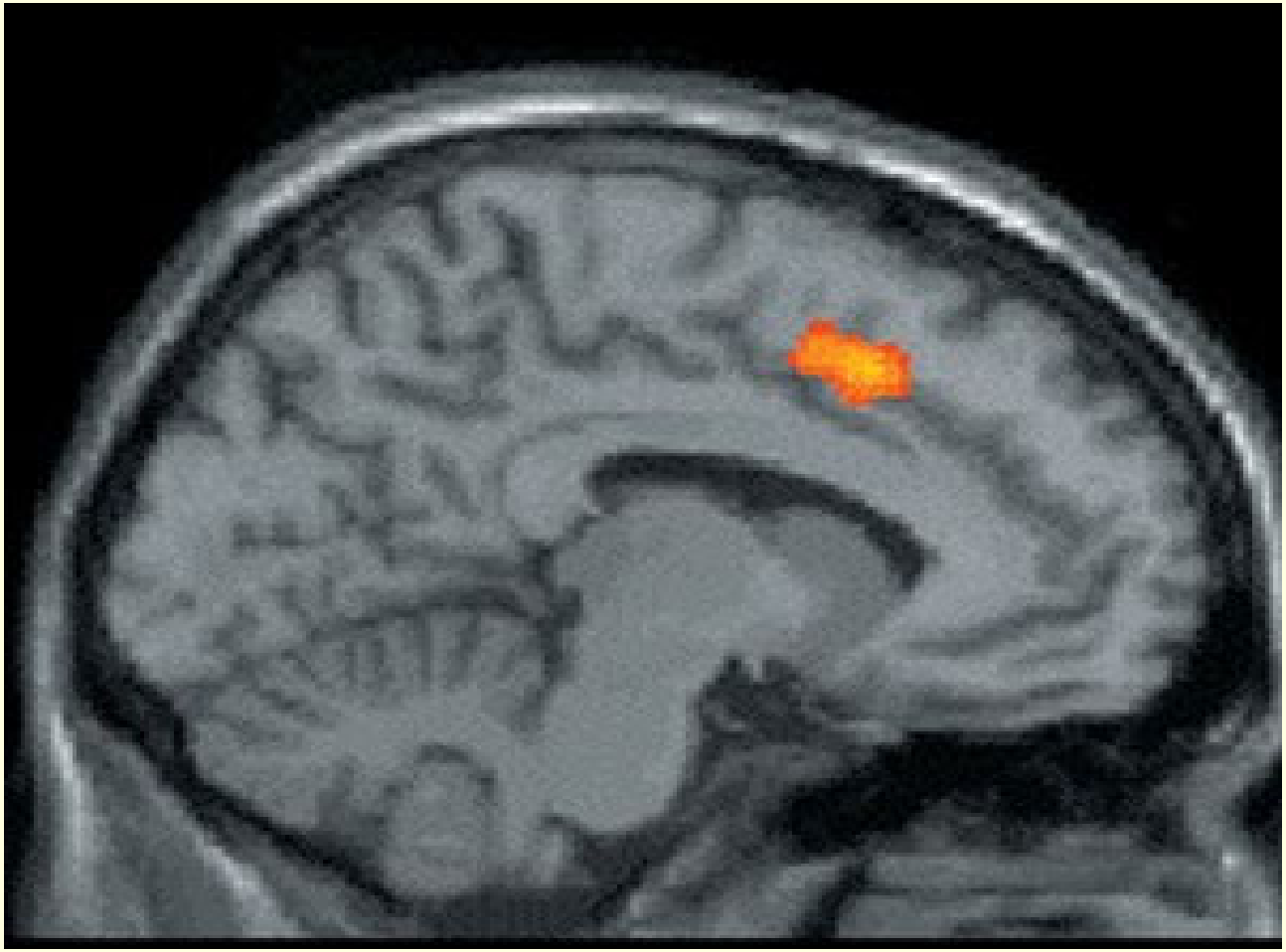
and wisdom to know the difference.

Benefits of Unilateral Virtue

- Simplifies things: all you have to do is just live by your own code, and others will do whatever they do.
- Feels good in its own right.
- Minimizes inflammatory triggers, evokes good treatment, empowers you to ask for it.
- Stands you on the moral high ground.

Remaining virtuous in the face of provocation is a profound expression of non-harming and benevolence toward all beings.

“Ardent, Diligent, Resolute, and Mindful”



Interdependent Causes Affecting All Beings

- Physical: Pain, age, temperament, intelligence
- Realities of daily life: Class, race, gender, job, responsibilities, stresses
- History: Childhood, major events as an adult
- Psychological: Personality, values, fears and hot buttons, hopes and dreams
- Their parents, and the factors that shaped their parents
- Their grandparents, great-grandparents . . . back and back
- So many forces upstream, forming the river of causes flowing through our lives today . . .

Lunchtime Reflections . . .

Wisdom is . . . all about understanding the underlying spacious and empty quality of the person and of all experienced phenomena.

To attain this quality of deep insight, we must have a mind that is quiet and malleable.

Achieving such a state of mind requires that we first develop the ability to regulate our body and speech so as to cause no conflict.

Venerable Ani Tenzin Palmo

What Is Empathy?

- Sensing, feeling, and understanding how it is for the other person
- Both a bodily resonance and a conceptual understanding
- Empathy is usually communicated, though often tacitly.
- We can give empathy, we can receive it, and we can ask for it.

What Is to Be Empathized with?

- “Tracks” of experience – Sensations, “feeling tone,” emotions, thoughts, desires, images, memories

- Layers of the psyche:
  - Present-day, adult material on top, younger material beneath
  - Rational thought on top, rawer and more primal material beneath
  - Harsh, armored feelings like anger on top, softer feelings beneath
  - Problematic strategies on top, universal and wholesome deep needs beneath.

- Inclinations – Temperament. Personality. Mood. Psychological effects of health issues. History. Intelligence. Culture. Gender. Fatigue, hunger, irritability, ripples from recent events, etc.

- Dynamics – Wants pulling in multiple directions. Personal imperatives. Expression of true self pressing through defenses against just that.

Empathic Individuality

- Empathy involves joining . . .
- . . . But it does not require approval, agreement, or giving up your own wants.
- Separate enough to look inside and use your own responses as information
- Empathy is not required for virtue.
- Executive functions - the core of virtue - are needed for empathy.

Why Is Empathy Important?

- We benefit from both receiving empathy and giving it.
- We all need to feel *felt.* To exist as a Thou to their I.
- Empathy breakdowns are routine, and just need repairs.
- Without repairs, mistrust grows and relationships shrink
- The more vulnerable we are, and the higher the stakes, the greater the impacts of empathic breakdowns.
- Empathy says, “Message received.” Often, that’s all the situation requires, and if not, it’s still easier to deal with the topic at hand.
  - Empathy provides useful information.
  - Especially helpful with people who are aversive to you.

Empathy in Buddhist Practice

- Empathy expresses Wise View, which sees how we are all related to each other.
  - Empathy is thus, in a deep sense, self-understanding
- Empathy is sila in action, the restraint of reactive patterns to be present with others.
- Empathy involves non-attachment to view, dropping our “case,” entering into “don’t know mind.”
- Empathy embodies non-harming.
- Compassion is not enough; we need to feel the weight of the other person’s experience.

Contents of Experience

- Sensations . . . Feeling tone . . . Emotions . . . Thoughts . . . Desires . . . Images . . . Memories . . .
- Adult material and young material . . . Rational thought and primal surges . . . Hard, protected emotions and soft, vulnerable emotions . . . Problematic desires and wholesome needs . . .
- Temperament . . . Personality . . . Mood . . . Health issues . . . History . . . Culture . . . Gender . . . Fatigue, irritability, worry . . .
- Wants pulling in different directions . . . Personal imperatives, hot buttons . . . True self pushing through defenses against it . . .

Feeling Strong

- Relaxed, resting in awareness
- Feeling the strength in awareness itself
- Energy and strength in your breathing . . . in arms and legs . . . in your whole being . . .
- A spacious strength that lets others flow through
- In relationship and at peace

Empathy Skills (so far)

- Awareness of experience
- Openness to others
- Sustaining mindfulness to them

More Empathy Skills

- Reading emotion in face and eyes
- Dropping aversion (judgments, distaste, fear, anger, withdrawal)
- Inferential imagination
- Expressing empathic understanding
  - Reflecting the content
  - Resonating with the tone and implicit material
  - Questions are fine
  - Respect and wise speech throughout

Reflections about Empathy

- You're more likely to get empathy if you're:
  - Open, present
  - Honest, real, authentic
  - Reasonably clear
  - Responsible for your own experience
  - Taking it in when you feel felt

- Empathy can be negotiated:
  - Name it as a topic in the relationship
  - Follow NVC format: "When X happens, I feel Y, because I need Z. So I request _____ ."
  - Stay with it.

Grandchildren!



Neurological Building Blocks of Empathy

- Understanding the intentions of others
- Simulation systems:
  - Activate both when we are engaged with something, and when we see others engaged
  - Involving actions, emotions, pain, and more
  - Many circuits, including mirror neurons
  - Automatic, continual re-creation within us of traces of the experience of others

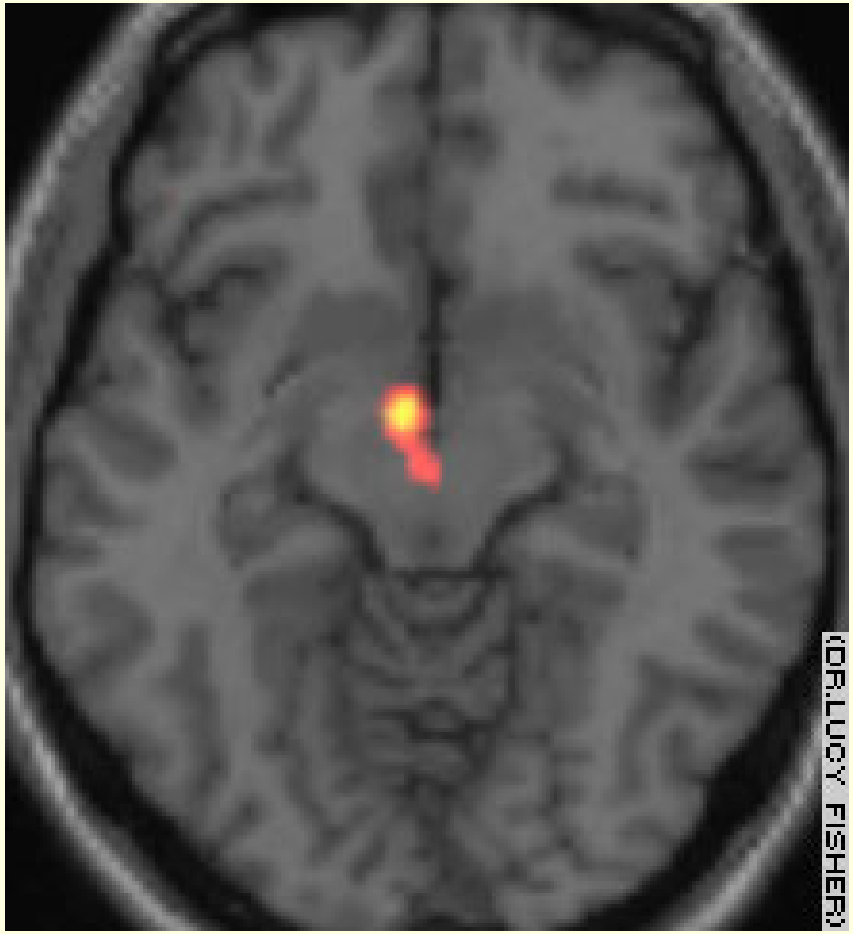
More Building Blocks of Empathy

- Cooperative planning:
  - Reproductively advantageous if reputation is known
  - Strong teamwork beats rival groups, especially in harsh environments
- Language:
  - Enables empathic inferences
  - Empathy is inquiry, not just resonance
- Conceptualization:
  - Lets us try on the perspective of others
  - Aids mental flexibility

Evolution of Bonding and Love

- Because childhood is so long - because humans are so smart
- Virtue and empathy promote bonding.
- Other factors as well, including biochemical processes involving dopamine and oxytocin

The Brain in Love



The Brain in Love and Lust

- Tegmentum floods caudate with dopamine; caudate signals: “Send more!”
- Dopamine involved in pleasure and motivation (cocaine triggers waves of it)
- Being rejected in love activates the insula, also involved in the experience of physical pain.
- Lust activates hypothalamus and amygdala: drives, emotion, and arousal
- Dopamine increases testosterone, and interest in sex

More grandchildren . . .



Oxytocin

- Promotes bonding between mothers and children, and between mates, keeping those kids alive
- In women, triggers let-down reflex in nursing, and tend-and-befriend behaviors during stress
- In both women and men it dampens stress response, and it feels pleasurable, relaxed, a “rightness”
- Rewards all bonding behaviors
- Triggered by:
  - Nipple stimulation
  - Orgasm
  - Physical contact (especially skin to skin)
  - Moving together harmoniously (dancing)
  - Warm feelings of rapport or love (lovingkindness); devotion
  - Imagination of these

Achy, Breaky, Shaky Love

- Rewards of mating can lead to looking for love in the wrong places, or heartbreak with rejection.

- Buddhist analysis (Chain of Dependent Origination):

Contact -> Feeling tone -> Craving -> Clinging -> Suffering

- Insight (Wise View):
  - Understands the slippery slope of desire
  - Sees, and accepts, impermanence
  - Fosters healthy disenchantment

- Equanimity:
  - Not reacting to our (feeling tone) reactions
  - Enjoying pleasures as they flow through, without grasping
  - Not adding the second dart to what's unpleasant
  - Patience with what's neutral

Us and Them

- With all the evolutionary pressures and rewards for empathy, cooperation, bonding, and love - why all the beastliness?
- The rewards of “us” naturally pull inward, away from “them.”
- Core evolutionary strategy: within-group cooperation, and between-group aggression.
- Both capacities hard-wired into our brains, ready for activation.
- Two wolves in your heart: Which one will you feed?
- Who is not in the room? Who is not held in your heart?

Feeding the Wolf of Love

- Sila/restraint about over-identifying with “us”
- Consider others as young children.
- Notice good things about neutral or unpleasant people.
- Focus on similarities between “us” and “them.”
- Consider others as your mother or dear friend in a past life.
- Reflect on the suffering of so many people in the world.
- Self-generate feelings of kindness and love.
- Bring to mind the sense of someone who cares about you.
- Keep extending out the sense of “us” to include everyone.

So that all cubs are our own . . .

So that all beings are our clan . . .

All life, our relatives . . .

The whole earth, our home . . .

May you know love, joy, wonder, and wisdom,
in this life, just as it is.

Thank you!